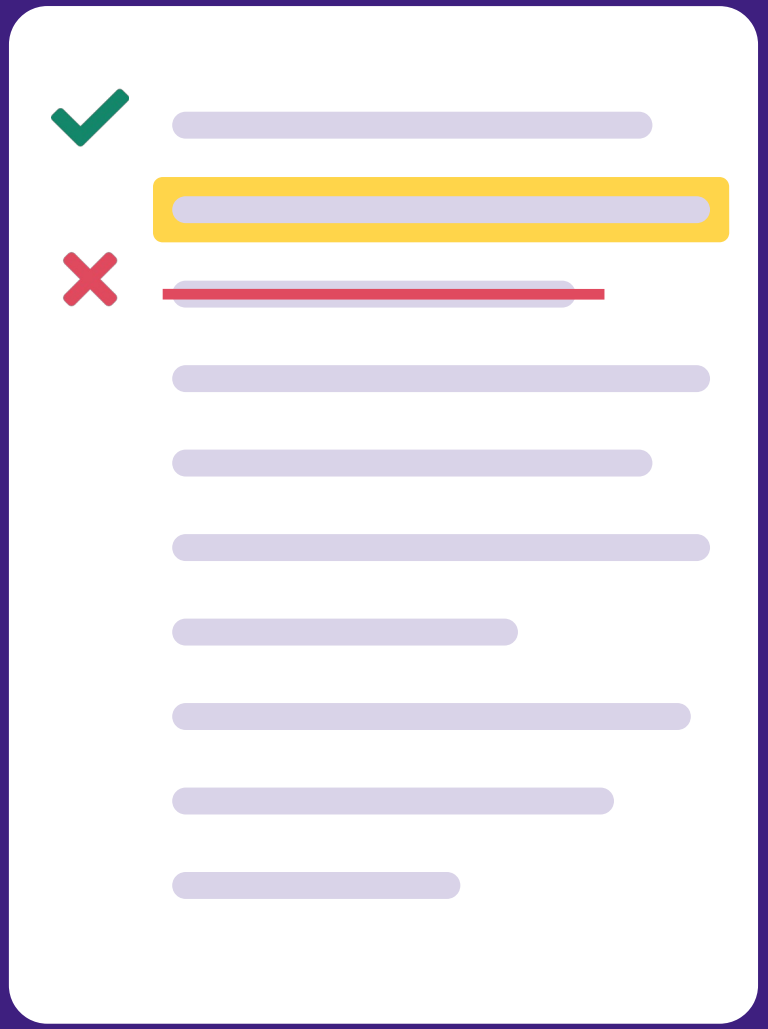
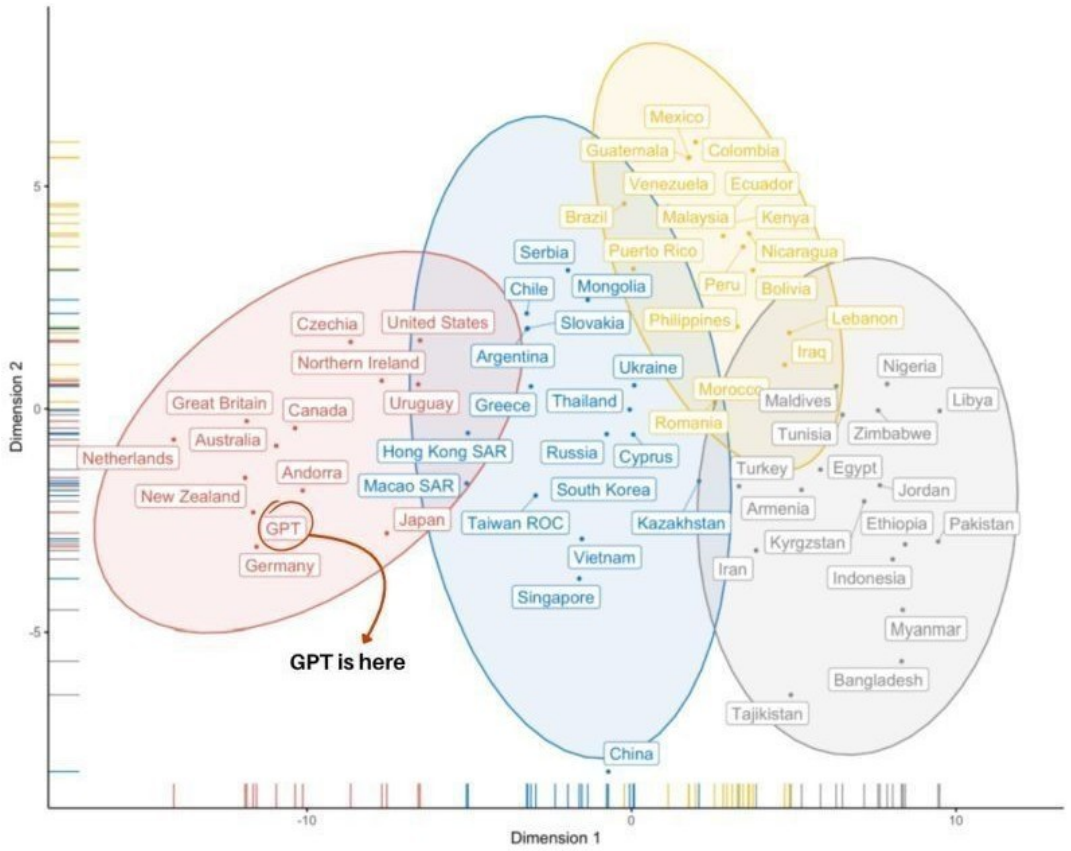
Enhancing Literacy and Numeracy Skills Using AI

Introduction to AI in Education: Professional Judgement and Purposeful Use



Ground

Where does AI's view of the world come from?



A large language model answered World Values Survey questions, and its answers were compared with people from 65 countries.

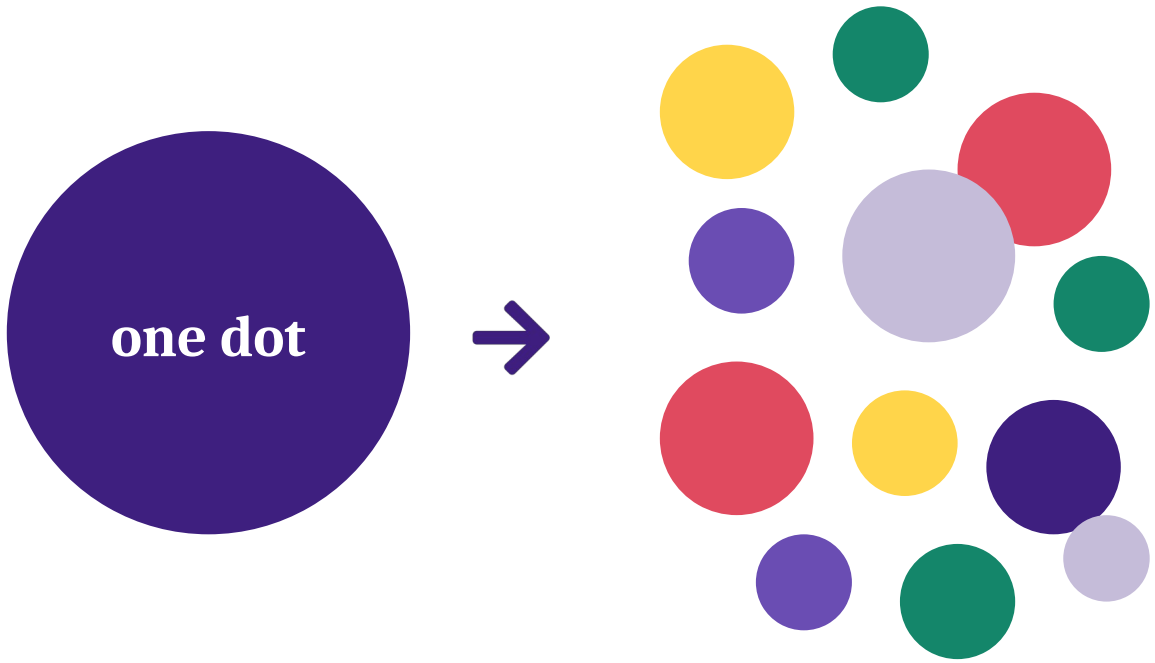
Its answers were closest to people from Western, educated, industrialised, rich and democratic (WEIRD) societies.

The further a country is culturally from the United States, the less the model's answers resemble its people's.

Atari et al., 2023

Ground

Now question the map itself



Each dot is a whole country, but Indonesia and Australia are not single cultures. One dot hides many regions, languages, communities and ways of life.

Whose view of the world does this map show, and whose does it leave out?

Ground

“What kind of teacher do you want your PPG students to become, and what should AI never take away?”

This program sits within Indonesia’s Pembelajaran Mendalam (Deep Learning) approach and national guidance that AI supports the teacher and does not replace them.

Kementerian Pendidikan Dasar dan Menengah, 2025; BSKAP, 2025

Readiness

Where is AI already touching your work?

“Where is AI already entering your work, directly or indirectly?”

- teaching
- mentoring
- teacher education
- administration
- resource preparation
- assessment

possible value

professional concern

Readiness

Why professional judgement matters

Request: "Explain to learners who can identify halves and quarters how to decide whether 3/4 or 2/3 is greater."

Response 1

To compare fractions, find a common denominator and then compare the numerators. The common denominator of 3/4 and 2/3 is 12. Rewrite 3/4 as 9/12 and 2/3 as 8/12. Because 9 is greater than 8, 3/4 is greater than 2/3. This method works for any two fractions.

Response 2

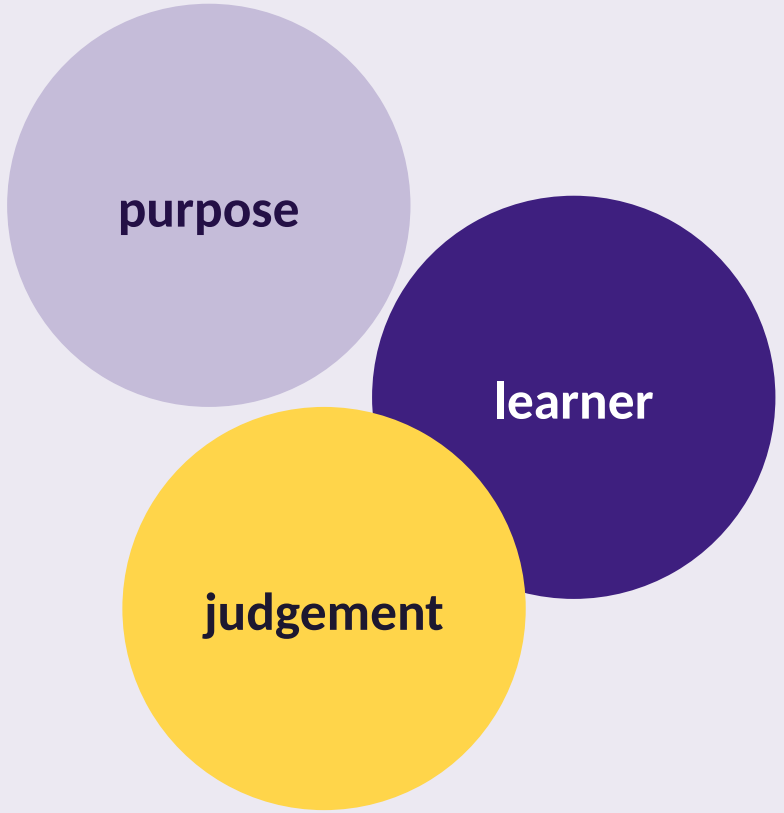
Let's compare 3/4 and 2/3 by cutting both into twelfths. Three quarters is the same as 9/12, and two thirds is the same as 6/12. Since 9/12 is bigger than 6/12, three quarters is the larger fraction. Great work comparing them!

Response 3

Cross-multiply the numerators and denominators: $3 \times 3 = 9$ and $2 \times 4 = 8$. Because $9 > 8$, $3/4 > 2/3$. In general, $a/b > c/d$ if and only if $ad > bc$, where b and d are positive integers.

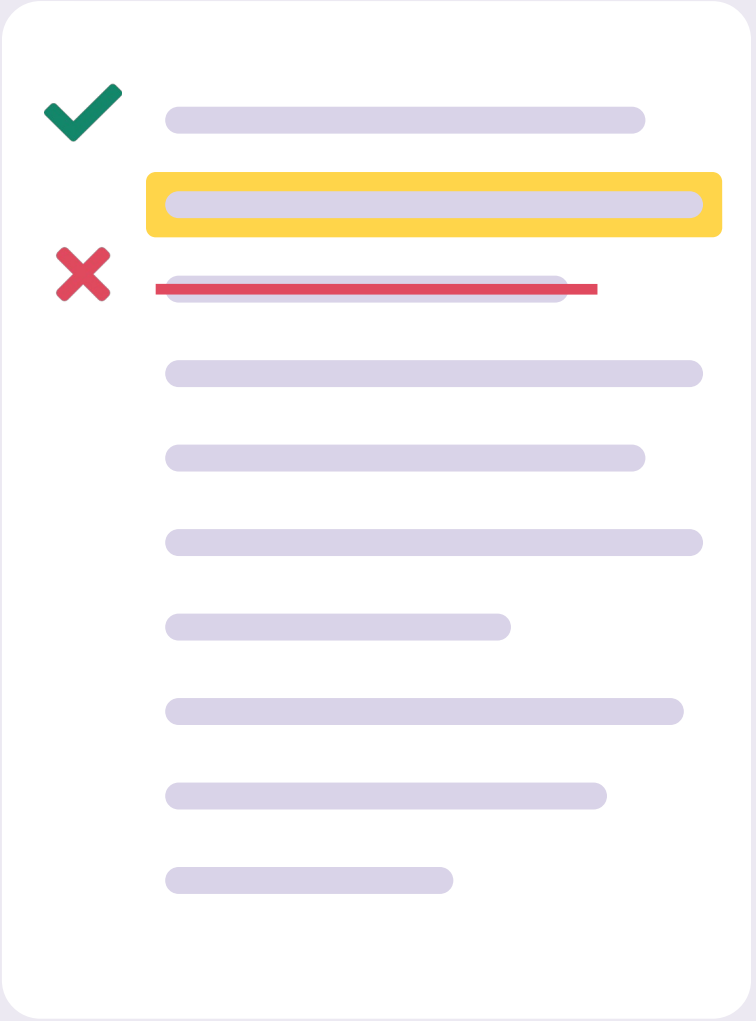
Which response is best?

“Best for what purpose, for which learner, and on whose judgement?”



Generative AI can produce convincing material quickly, but educators remain responsible for deciding whether that material is accurate, appropriate and educationally useful.

(Chiu, 2025; Memarian & Doleck, 2024; Miao & Holmes, 2023)



Readiness

We will now work through three things

Concept 1

What the system is doing

Concept 2

How our instructions shape what it gives us

Concept 3

How we decide whether an educational use is worth keeping

Concept 1

Diagnose the resource together

AI-generated resource

Learning goal: Learners will compare fractions.

To add 1/2 and 1/4, add the tops and add the bottoms: 1/2 + 1/4 = 2/6. Research from the Hartwell Institute (2021) shows that every learner masters fractions after three lessons.

Activity: Cut a pizza into 8 slices, then measure each slice with a ruler marked in 1/16 inches.

Practice: 1/3 + 1/3 = 2/6 (Answer: 2/6) 1/2 + 1/4 = 2/6 (Answer: 2/6)

Diagnose it: what is wrong, weak or missing?

Compare your diagnosis with what others post in the chat.

Nominate one example where fluency could easily be mistaken for quality.

Concept 1

What is happening here?

Generative AI predicts plausible output from learned patterns.

It does not independently:

- ✕ verify truth
- ✕ understand curriculum intent
- ✕ know the learner

! **Hallucination:** confident generation of unsupported or incorrect content

Chiu, 2024; Chiu, 2025

Concept 1

Human-in-the-loop

AI may generate or suggest, but the educator decides what is accepted, revised or rejected.



Accepted



Revised



Rejected

Memarian & Doleck, 2024; Miao & Holmes, 2023

Concept 1

Appropriate is not the same as useful for learning.

Correct is not the same as appropriate.

Fluent is not the same as correct.

Concept 2

Design for a context

Same brief for everyone: “Create an activity to help learners understand fractions.”

Context card A

Learners who can identify halves and quarters. 25 minutes. Only a whiteboard.

Context card B

Learners who can compare simple fractions but struggle to explain their reasoning in writing. English is an additional language.

Context card C

Preservice teachers who must teach fractions next week and feel unsure of their own content knowledge.

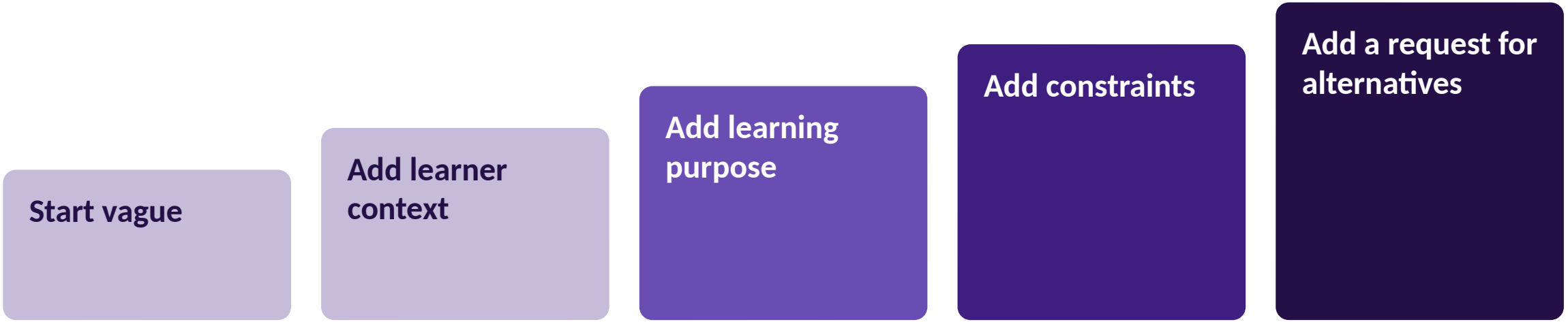
Context card D

A mixed-readiness class of 30 learners with one teacher. Some need support and some are ready for extension.

Design a prompt for your card and run it. Post the output without the card, and infer which professional decisions were or were not communicated in someone else’s.

Concept 2

Clearer prompts do not remove judgement



At each stage, ask: what improved and what risk remained?

Concept 2

A practical prompting pattern

Context

Who are the learners and what is happening?

Purpose

What do you want the material to help achieve?

Constraints

What must the response preserve, avoid or include?

Review

What will you check before deciding whether to use it?

Prompting is not a technical trick. It is the process of making professional intent explicit to the system.

Concept 2

“What did the prompt improve, and what could it never guarantee?”

Better prompting improves relevance. It does not transfer professional responsibility to the system.

Concept 3

Make the decision

Case: A teacher has 34 learners' written reflections to respond to by Friday. Several learners write in a mix of languages, and two have additional learning needs. The teacher is considering using AI to draft feedback on all of them.

Should AI be used at all?

Where does it enter the process?

What data can be used?

What must remain human?

What would you revise or reject?

Then share and defend your decision.

Concept 3

Judge the educational value, not just the output

Purpose

Does this actually address the educational problem?

Accuracy

Is the content correct and defensible?

Learner

Is it appropriate for these learners, at this point in their learning?

Context

Does it make sense in this cultural, institutional and classroom context?

Safety

Have I avoided sharing information that should not be given to the system?

Learning

Does the AI support learning, or has it done the important thinking for the learner?

Judgement

What must I change, verify or reject before use?

Revise or reject at least one part of your decision.

Analyse your decision using the seven lenses.

Safety: Mouta et al., 2024; Miao & Holmes, 2023. Learning: Chiu, 2024; Xia et al., 2024.

Concept 3

Four boundaries



AI can support differentiation and access, but changing the pathway should not automatically change the learning goal.



AI can help generate questions, exemplars and draft feedback, but it should not replace professional evaluation or grading.



AI can surface possibilities and patterns, but educators interpret meaning in context.



Learner information should not be shared simply because names have been removed. Re-identification risk still matters.

Assessment: Xia et al., 2024. Educator oversight: Memarian & Doleck, 2024. Privacy: Mouta et al., 2024; Miao & Holmes, 2023; Rocher et al., 2019.

Concept 3

The learning goal is not “get a good AI answer”, but “make a defensible teaching decision”.

Express

What does this mean for my practice?

Think of a task from your own work. Complete six short statements:

My context

The teaching or teacher-education situation I am thinking about.

My purpose

What I want to improve for learners.

The AI role

What I might ask AI to contribute.

The human role

What requires my professional judgement.

My boundary

What I would not delegate or what information I would not provide.

My next step

What I want to test or develop during the rest of the programme.

Express

Respond and close

Respond to a peer - 6 minutes

Read another participant's statements and respond with one question, one possible risk, or one way the idea could be strengthened.

Share - 4 minutes

Invite 2-3 participants to share what changed in their thinking.

This six-part statement is the first version of your AI-supported teaching decision, which later modules will refine.

“The aim is not to use AI more. It is to make better teaching decisions about when, why and how to use it.”

As the programme develops, we will return to these boundaries in greater depth, particularly questions of bias, cultural and linguistic representation, privacy, equity of access, transparency and responsible use.

References

Atari, M., Xue, M. J., Park, P. S., Blasi, D. E., & Henrich, J. (2023). Which humans? PsyArXiv. https://doi.org/10.31234/osf.io/5b26t

Badan Standar, Kurikulum, dan Asesmen Pendidikan. (2025). Panduan pemanfaatan kecerdasan artifisial untuk guru. Kementerian Pendidikan Dasar dan Menengah. https://repositori.kemendikdasmen.go.id/36443/

Chiu, T. K. F. (2024). The impact of generative AI (GenAI) on practices, policies and research direction in education: A case of ChatGPT and Midjourney. Interactive Learning Environments, 32(10), 6187-6203. https://doi.org/10.1080/10494820.2023.2253861

Chiu, T. K. F. (2025). AI literacy and competency: Definitions, frameworks, development and future research directions. Interactive Learning Environments, 33(5), 3225-3229. https://doi.org/10.1080/10494820.2025.2514372

Kementerian Pendidikan Dasar dan Menengah. (2025). Naskah akademik pembelajaran mendalam: Menuju pendidikan bermutu untuk semua.

Memarian, B., & Doleck, T. (2024). Human-in-the-loop in artificial intelligence in education: A review and entity-relationship (ER) analysis. Computers in Human Behavior: Artificial Humans, 2(1), 100053. https://doi.org/10.1016/j.chbah.2024.100053

Miao, F., & Holmes, W. (2023). Guidance for generative AI in education and research. UNESCO. https://doi.org/10.54675/EWZM9535

Mouta, A., Pinto-Llorente, A. M., & Torrecilla-Sanchez, E. M. (2024). Uncovering blind spots in education ethics: Insights from a systematic literature review on artificial intelligence in education. International Journal of Artificial Intelligence in Education, 34(3), 1166-1205. https://doi.org/10.1007/s40593-023-00384-9

Rocher, L., Hendrickx, J. M., & de Montjoye, Y.-A. (2019). Estimating the success of re-identifications in incomplete datasets using generative models. Nature Communications, 10, Article 3069. https://doi.org/10.1038/s41467-019-10933-3

Xia, Q., Weng, X., Ouyang, F., Lin, T. J., & Chiu, T. K. F. (2024). A scoping review on how generative artificial intelligence transforms assessment in higher education. International Journal of Educational Technology in Higher Education, 21, 40. https://doi.org/10.1186/s41239-024-00468-z